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Subject: Programming with Java

ASSIGNMENT No : 2

TITLE : Constructors in Java

Problem Statement

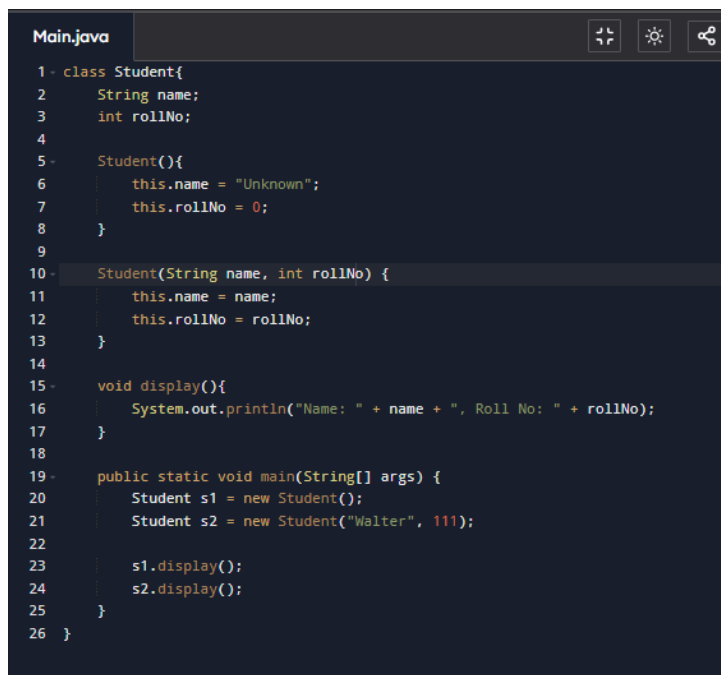
Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Programme

1. Create a Student class using default and parameterized constructors to initialize student name and roll number.



```
Main.java
1 class Student{
2     String name;
3     int rollNo;
4
5     Student(){
6         this.name = "Unknown";
7         this.rollNo = 0;
8     }
9
10    Student(String name, int rollNo) {
11        this.name = name;
12        this.rollNo = rollNo;
13    }
14
15    void display(){
16        System.out.println("Name: " + name + ", Roll No: " + rollNo);
17    }
18
19    public static void main(String[] args) {
20        Student s1 = new Student();
21        Student s2 = new Student("Walter", 111);
22
23        s1.display();
24        s2.display();
25    }
26 }
```

2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.

```
MobilePhone.java
1 class MobilePhone {
2     String brand;
3     String model;
4     double price;
5
6     MobilePhone(){
7         this.brand = "Generic";
8         this.model = "Basic";
9         this.price = 0.0;
10    }
11
12    MobilePhone(String brand, String model, double price){
13        this.brand = brand;
14        this.model = model;
15        this.price = price;
16    }
17
18    MobilePhone(MobilePhone phone){
19        this.brand = phone.brand;
20        this.model = phone.model;
21        this.price = phone.price;
22    }
23 }
```

```
18 MobilePhone(MobilePhone phone){
19     this.brand = phone.brand;
20     this.model = phone.model;
21     this.price = phone.price;
22 }
23
24 void display() {
25     System.out.println("Brand: " + brand + " | Model: " + model + " | Price: Rs " +
26         price);
27 }
28
29 public static void main(String[] args) {
30     MobilePhone p1 = new MobilePhone("Apple", "iPhone 17", 79999);
31     MobilePhone p2 = new MobilePhone(p1);
32     p1.display();
33     p2.display();
34 }
```

Output

Program 1 output:

```
Output
Name: Unknown, Roll No: 0
Name: Walter, Roll No: 111

=== Code Execution Successful ===
```

Program 2 output:

```
Output
Brand: Apple | Model: iPhone 17 | Price: Rs 79999.0
Brand: Apple | Model: iPhone 17 | Price: Rs 79999.0

=== Code Execution Successful ===
```

Conclusion

These programs demonstrate Java constructor overloading and object cloning techniques. The Student class illustrates how default and parameterized constructors allow objects to be initialized with either placeholder or custom data. The Mobile Phone system builds on this by using a copy constructor, which provides a clean and safe way to duplicate existing object records without manual property copying.